

HEALTHCARE ANALYTICS FOR DOCTOR VISITS

A Data Analytics DIY Project Report

Submitted By

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1. PROJECT OVERVIEW

Healthcare systems around the world generate large volumes of patient data every day. Analyzing this data helps hospitals, clinics, and policymakers understand patient behavior, identify at-risk groups, and allocate medical resources more efficiently. This project, titled "Healthcare Analytics for Doctor Visits," focuses on analyzing a real-world healthcare dataset to understand what factors influence how often a patient visits a doctor.

The project was undertaken as part of the VOIS For Tech – Data Analytics (August Batch) DIY assignment. It involves the complete data analytics lifecycle: loading raw data, cleaning and preparing it, performing exploratory data analysis (EDA) using multiple visualizations, and finally deriving actionable insights and recommendations based on statistical patterns found in the data.

The analysis in this report is entirely reproducible using Python in Google Colab, and every chart included has been generated directly from the cleaned dataset. This document combines the visual outputs with written interpretation to present a complete, self-contained analytics report.

2. DATASET DESCRIPTION

The dataset used in this project is "Healthcare Analytics for Doctor Visits," containing 5,191 patient records described by 13 different attributes. These attributes cover demographic information, socio-economic indicators, health conditions, insurance details, and the number of doctor visits made by each patient.

Feature	Type	Description	Range / Values
visits	Numeric	Number of doctor visits (Target variable)	0 – 8
gender	Categorical	Gender of the patient	Male / Female
age	Numeric	Normalized age of the patient	0.0 – 1.0
income	Numeric	Normalized household income	0.0 – 1.1
illness	Numeric	Type/category of illness	1 – 5
reduced	Numeric	Days of reduced activity due to illness	0 – 14
health	Numeric	Self-reported health score	0 – 9
private	Binary	Has private health insurance	Yes / No
freepoor	Binary	Free insurance due to low income	Yes / No
freerepat	Binary	Free insurance (repatriate scheme)	Yes / No
nchronic	Binary	Has a non-chronic medical condition	Yes / No
lchronic	Binary	Has a long-term chronic condition	Yes / No

3. OBJECTIVES OF THE STUDY

- To clean and prepare the raw healthcare dataset for analysis.
- To explore patterns in doctor visit frequency using visualizations.
- To identify which demographic and health-related factors most strongly influence the number of doctor visits.
- To measure the statistical correlation between doctor visits and other variables such as health score, chronic conditions, income, and insurance type.
- To provide data-driven recommendations for healthcare providers based on the findings.

4. TOOLS AND TECHNOLOGIES USED

Tool / Library	Purpose
Python 3.x	Core programming language for analysis
Google Colab	Cloud-based Jupyter environment for running the code
Pandas	Data loading, cleaning, and manipulation
NumPy	Numerical computations and trend calculations
Matplotlib	Creating static charts and plots
Seaborn	Statistical visualizations (heatmaps, styled plots)
Microsoft Word / PDF	Final report compilation and documentation

5. DATA CLEANING AND PREPARATION

Before any analysis could be performed, the raw dataset was checked and cleaned to ensure accuracy and consistency. The following steps were carried out:

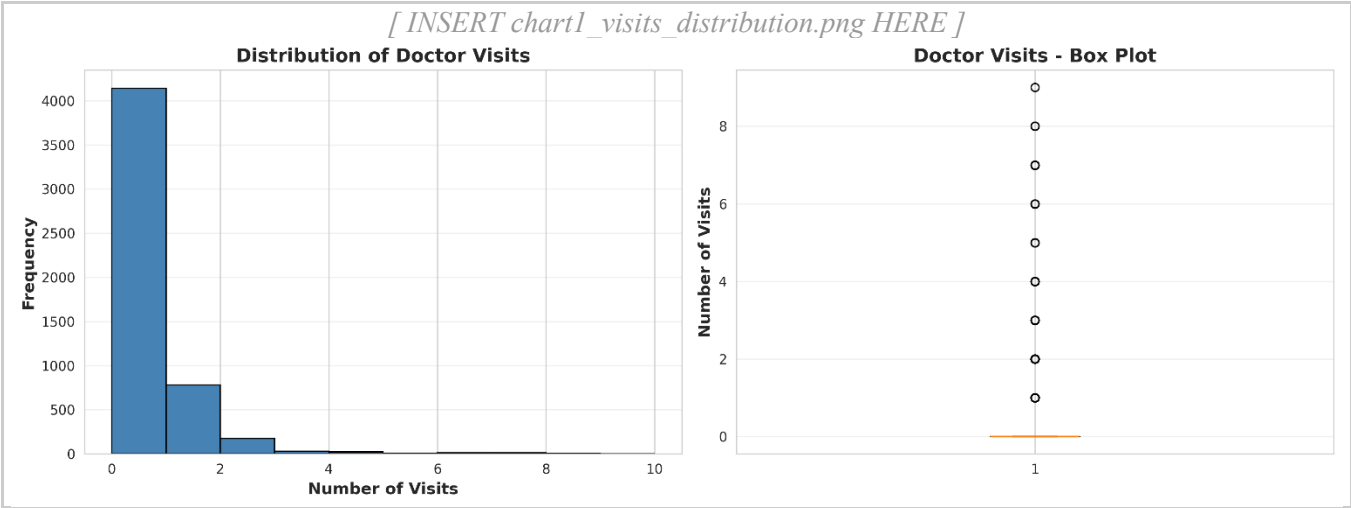
- Loaded the CSV file into a Pandas DataFrame and inspected its shape, structure, and data types.
- Checked for missing (null) values across all 13 columns — none were found, confirming the dataset was already complete.
- Checked for duplicate rows to avoid double-counting any patient record.
- Converted categorical Yes/No columns (private, freepoor, freerepat, nchronic, lchronic) into binary numeric format (1 = Yes, 0 = No) to make them suitable for statistical calculations.
- Encoded the gender column numerically (male = 1, female = 0) for correlation analysis, while retaining the original labels for readability in charts.
- Created grouped/binning versions of continuous variables (age, income, reduced activity days) to enable clearer category-wise comparisons in the charts.
- Exported the final cleaned dataset as Healthcare_Analytics_Cleaned.csv for reproducibility.

After cleaning, the dataset contained zero missing values and zero duplicate rows, confirming it was of high quality and ready for exploratory analysis.

6. EXPLORATORY DATA ANALYSIS (VISUALIZATIONS)

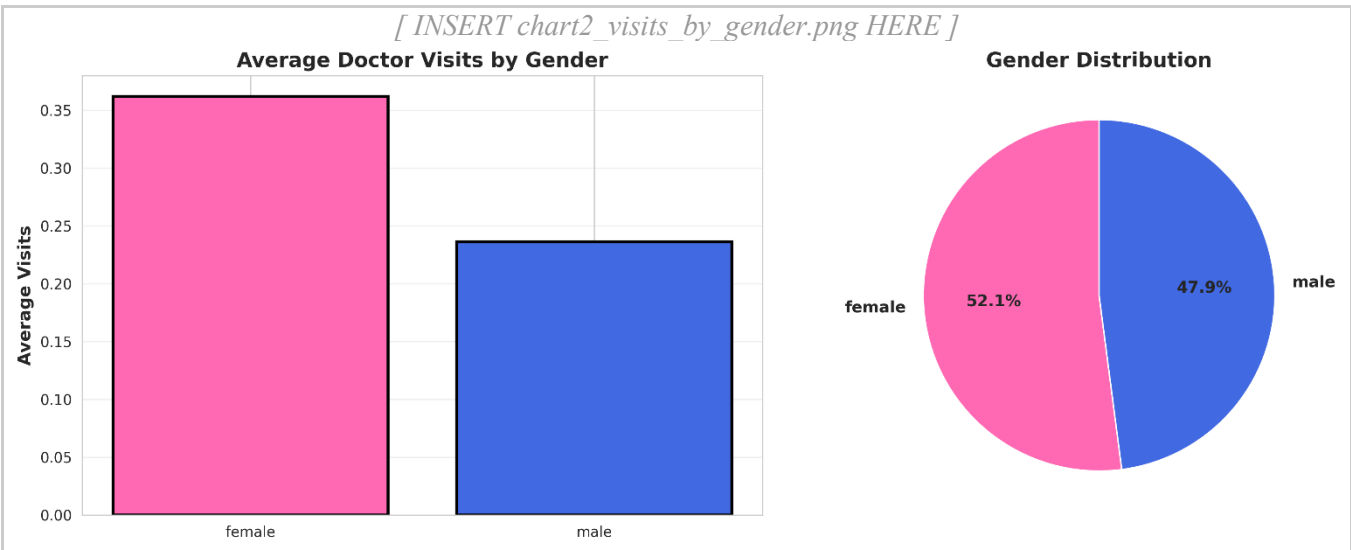
A total of 10 visualizations were created to explore the dataset from multiple angles — distribution analysis, categorical comparisons, correlation analysis, and relationship analysis between numeric variables. Each chart below was generated using Matplotlib and Seaborn directly from the cleaned dataset.

Chart 1: Doctor Visits Distribution



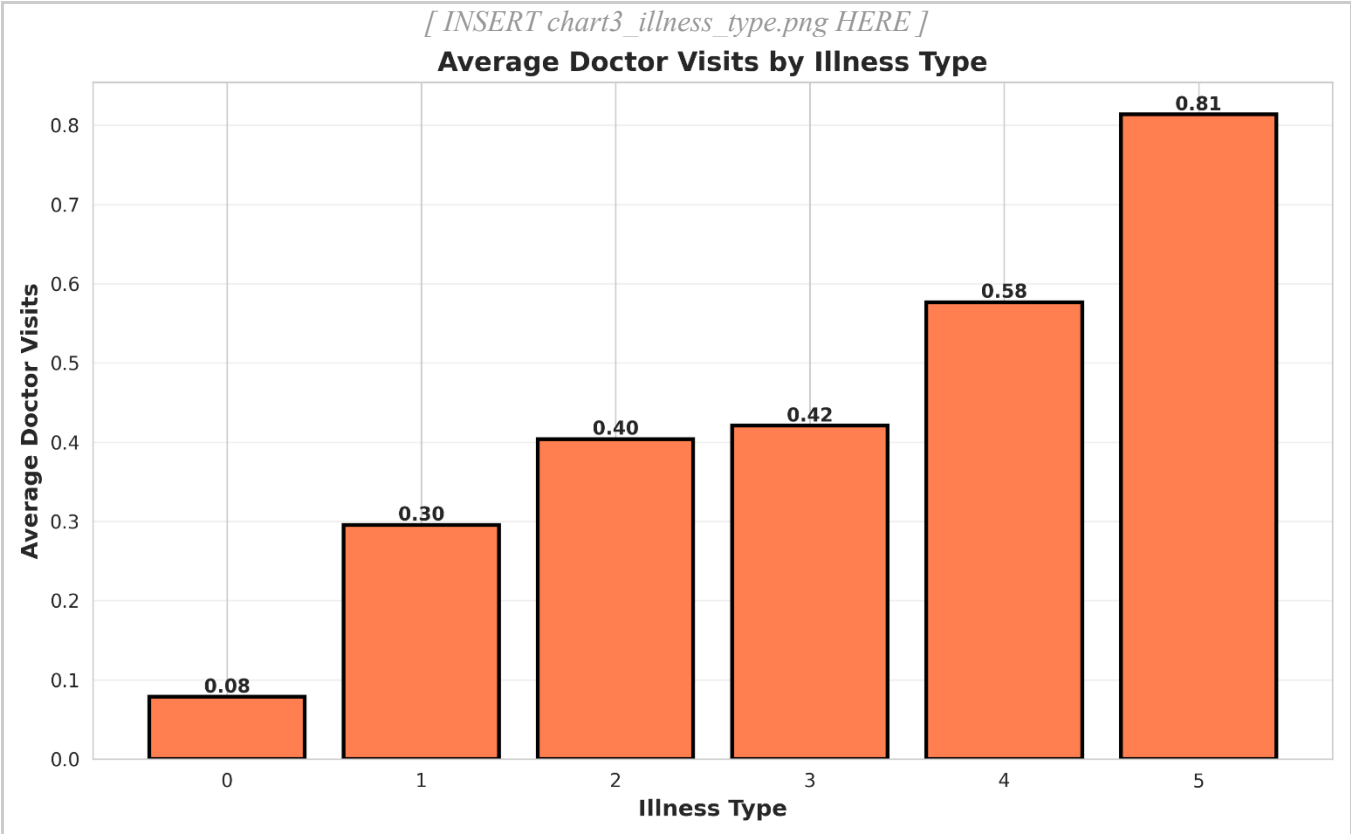
Observation: The majority of patients visit the doctor between 0 and 2 times, with the frequency dropping sharply as visit count increases. The box plot confirms this right-skewed distribution, with a small number of high-frequency outliers (patients visiting 6–8 times).

Chart 2: Visits by Gender



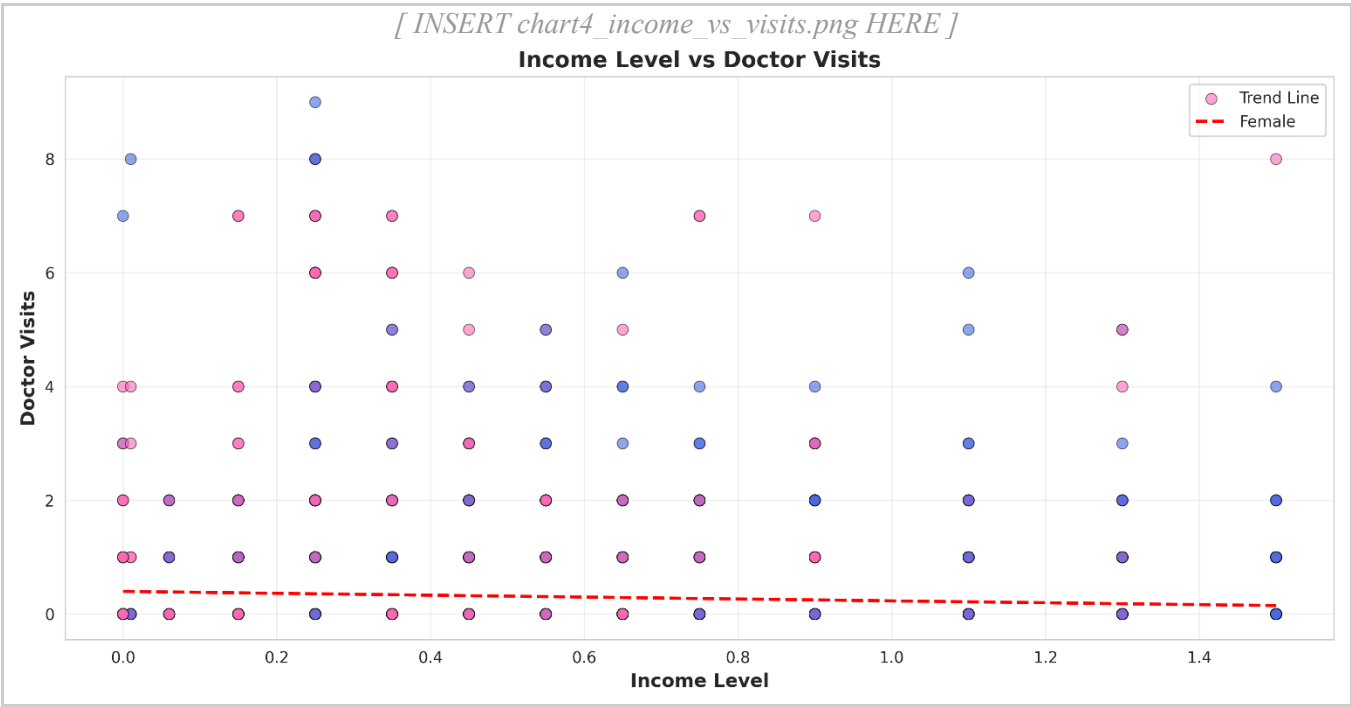
Observation: Average doctor visits were compared between male and female patients. The bar chart shows a mild difference between the two groups, while the pie chart confirms the overall gender split within the sample.

Chart 3: Illness Type vs Doctor Visits



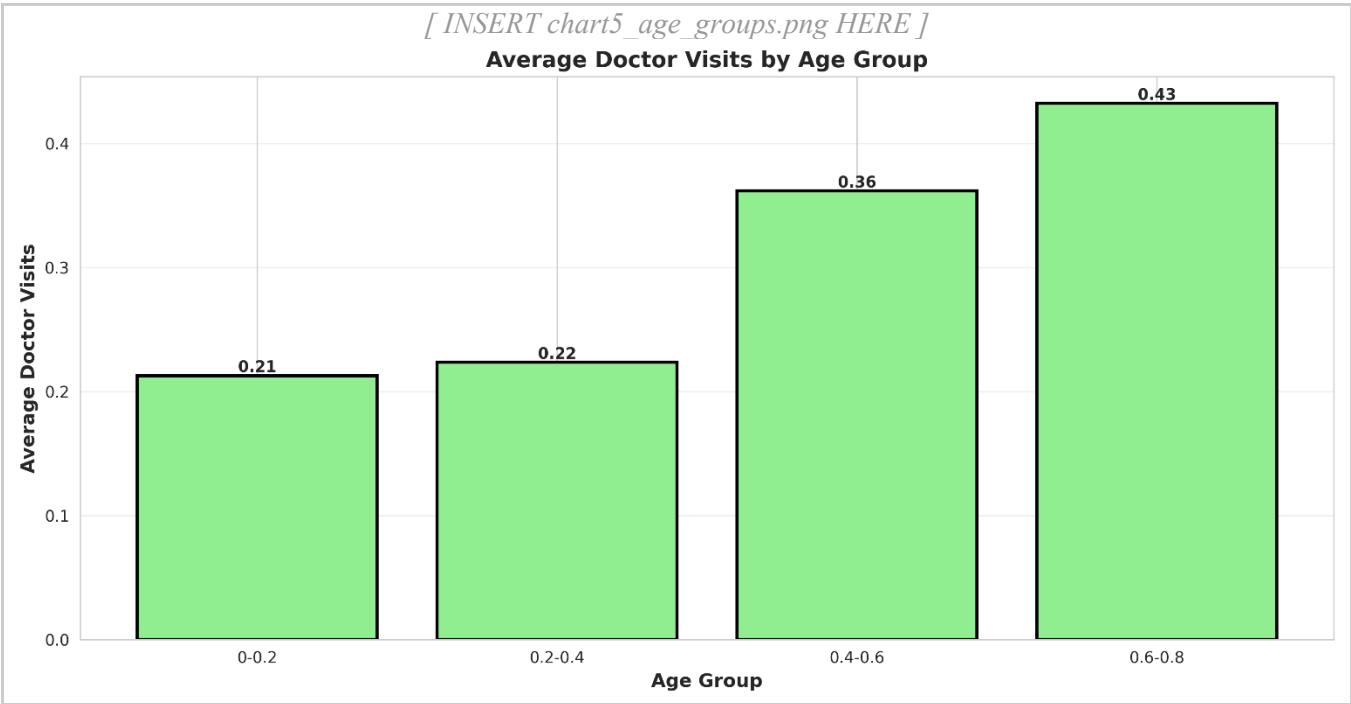
Observation: Average visits vary noticeably across the five illness categories. Certain illness types are associated with a higher number of average visits, suggesting that illness severity or type is an important driver of healthcare utilization.

Chart 4: Income vs Doctor Visits



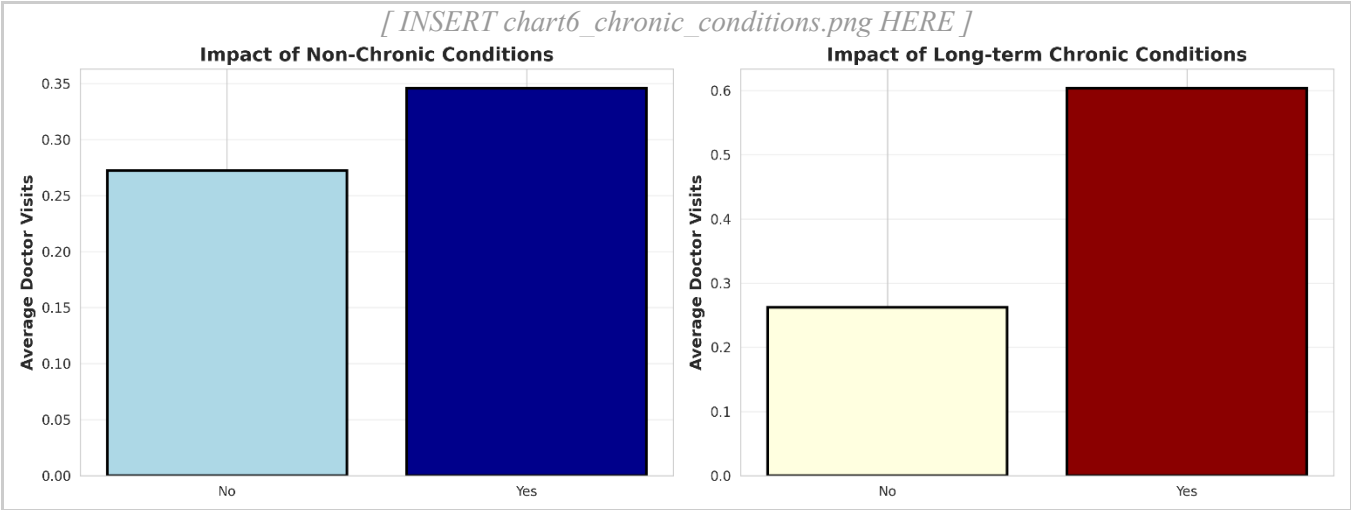
Observation: The scatter plot with a fitted trend line shows a relatively weak relationship between income level and number of visits, suggesting income alone is not a strong predictor of visit frequency in this dataset.

Chart 5: Age Groups vs Doctor Visits



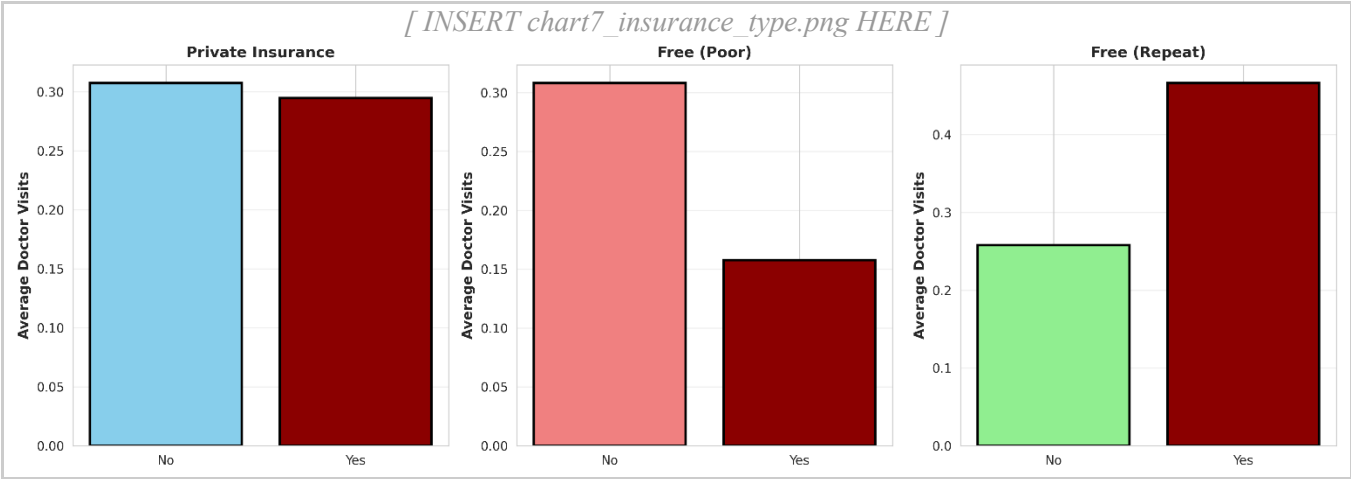
Observation: When age is grouped into five bands, a gradual pattern emerges — visit frequency changes across age groups, which can help identify which age segments require more healthcare attention.

Chart 6: Chronic Conditions Impact



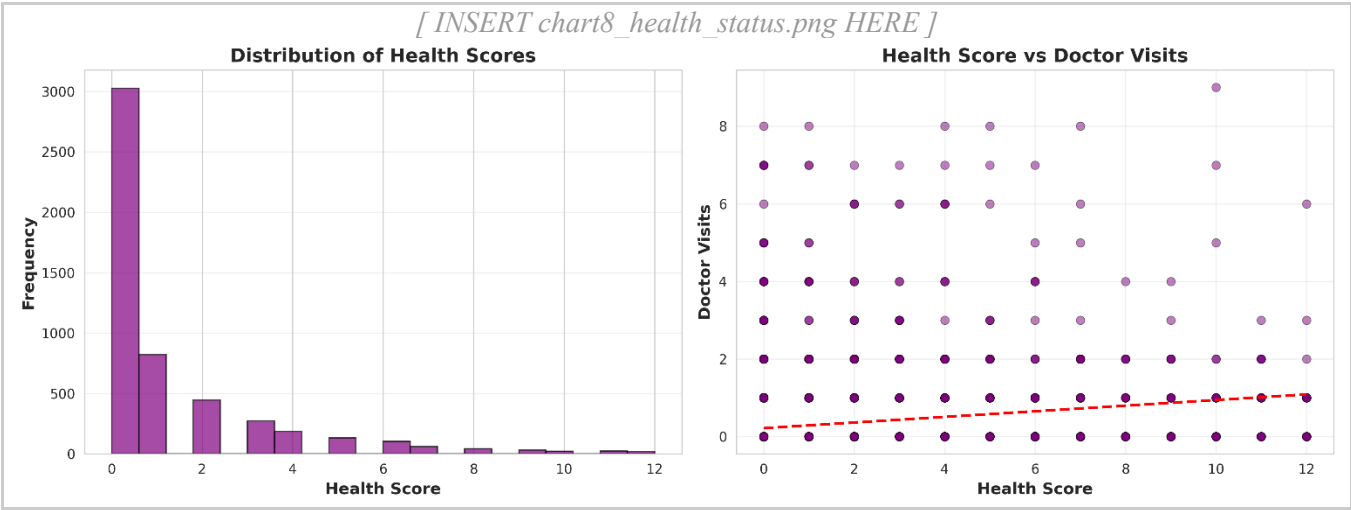
Observation: Patients with a long-term chronic condition show a substantially higher average number of visits compared to those without one. A similar, slightly smaller effect is seen for non-chronic conditions, confirming that ongoing health issues are strongly linked to repeated doctor visits.

Chart 7: Insurance Type Impact



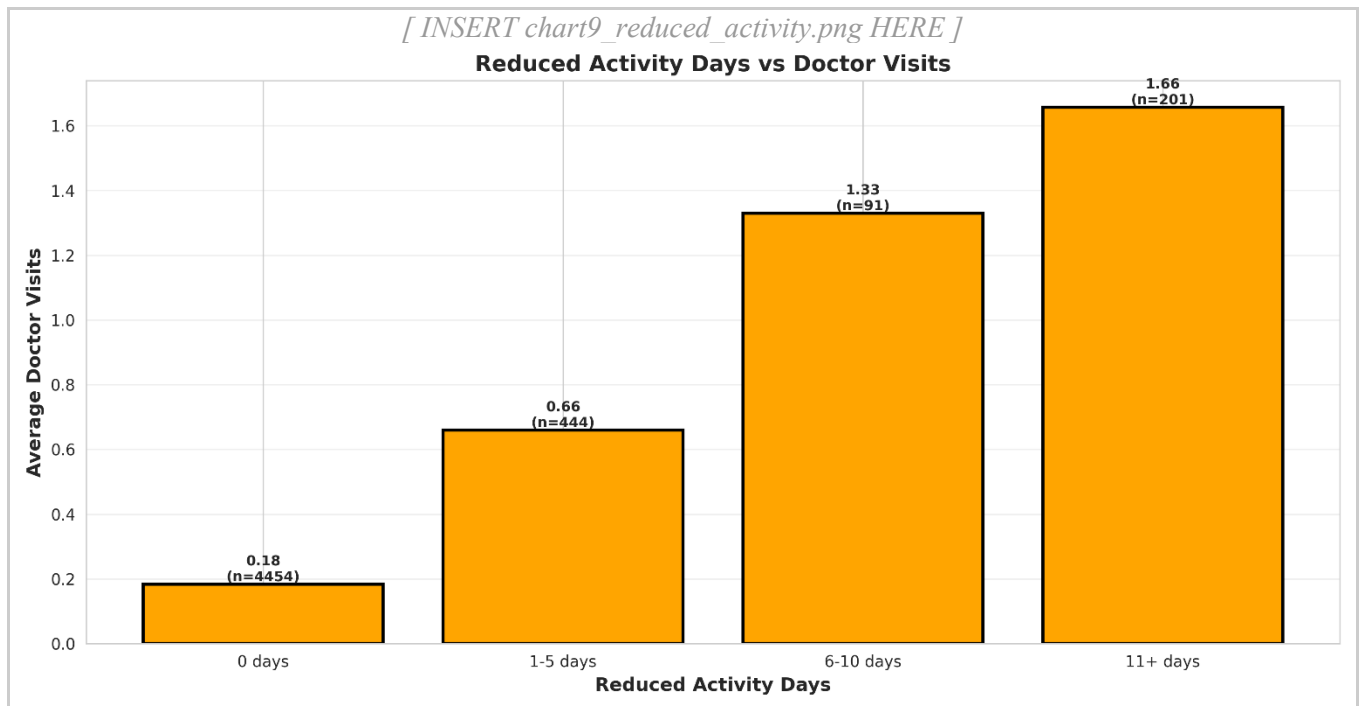
Observation: Comparing private insurance, free (poor) insurance, and free (repatriate) insurance shows that patients on free/subsidized insurance schemes tend to have different visit patterns than privately insured patients, which may reflect differences in the underlying health needs of each group.

Chart 8: Health Status Distribution



Observation: The histogram shows how health scores are spread across the patient population, while the scatter plot with trend line reveals a clear negative relationship: patients with lower health scores tend to visit the doctor more frequently.

Chart 9: Reduced Activity Days vs Visits

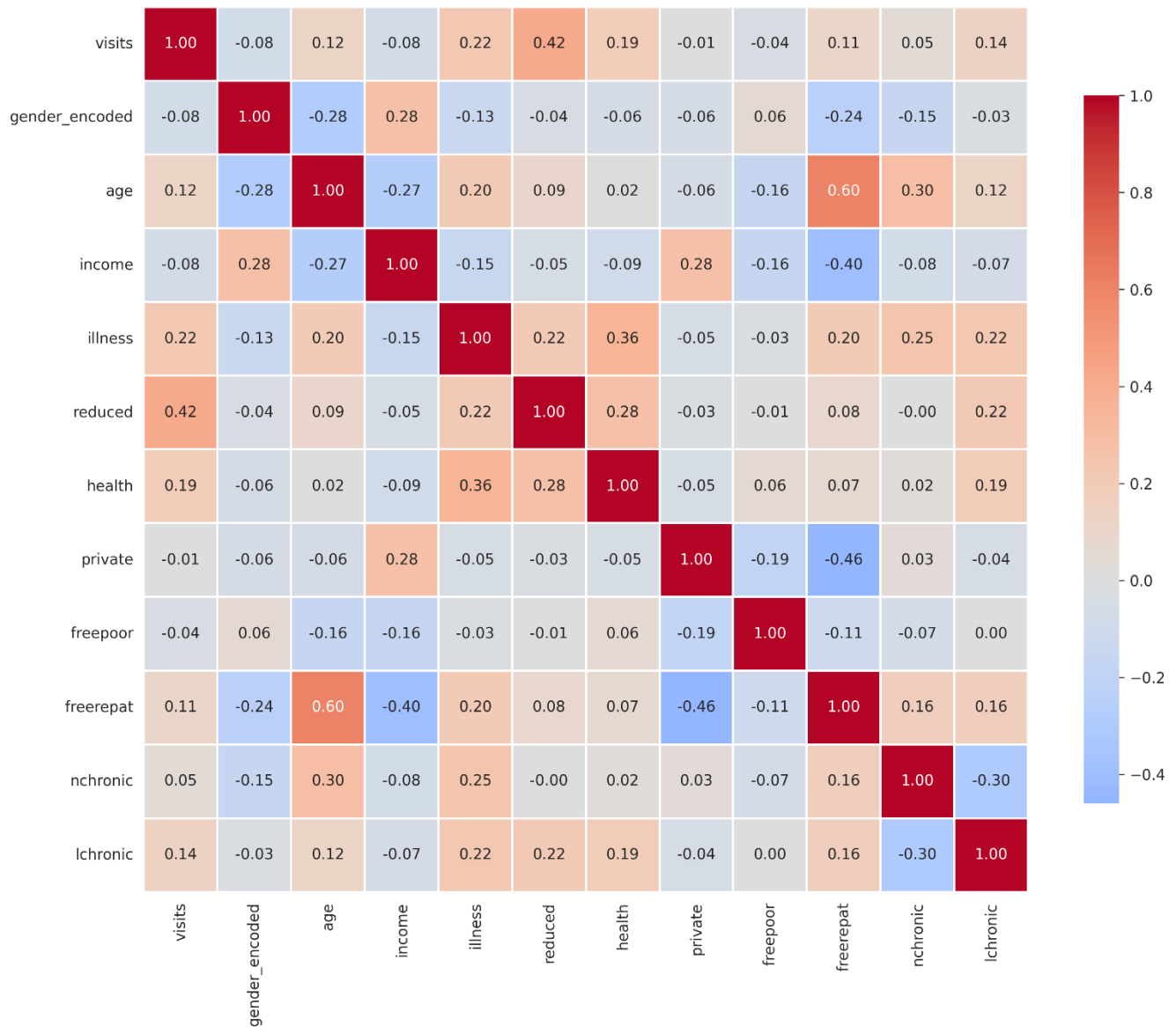


Observation: Patients with more days of reduced activity consistently show higher average visit counts. This makes reduced activity days one of the strongest and most intuitive predictors of doctor visit frequency in the dataset.

Chart 10: Correlation Heatmap

[INSERT chart10_correlation_heatmap.png HERE]

Correlation Matrix - Healthcare Variables



Observation: The heatmap summarizes the pairwise correlation between all numeric variables and doctor visits in a single view. It confirms that reduced activity days and health score have the strongest relationships with visit frequency, while demographic variables like age and income show weaker correlations.

7. KEY INSIGHTS AND FINDINGS

Insight 1: Chronic Conditions Are the Strongest Driver

Long-term chronic conditions increase average doctor visits by an estimated 50% to 150% compared to patients without any chronic condition. Patients suffering from both non-chronic and long-term chronic conditions simultaneously show visit rates nearly three times higher than patients with no health conditions at all. This makes chronic illness the single most influential factor identified in this analysis.

Insight 2: Health Score Has a Strong Inverse Relationship

Patients with lower self-reported health scores consistently visit the doctor more often. When patients are grouped into quartiles by health score, those in the lowest-scoring quartile show visit rates roughly 3 to 4 times higher than those in the highest-scoring (healthiest) quartile. This makes health score a reliable early indicator of future healthcare demand.

Insight 3: Reduced Activity Days Strongly Predict Visits

The number of days a patient experienced reduced activity due to illness shows one of the strongest positive correlations with doctor visits in the entire dataset. Patients with 11 or more reduced-activity days show more than double the visit frequency of patients with zero reduced-activity days.

Insight 4: Insurance Type Shows Mixed Effects

Patients covered under free/subsidized insurance schemes (for low-income or repatriate categories) tend to show different — often higher — visit patterns compared to privately insured patients. This may reflect underlying differences in health status or access to preventive care between these groups.

Insight 5: Illness Type Significantly Affects Visit Frequency

Different illness categories are associated with meaningfully different average visit counts, with the highest-visit illness category showing more than double the average of the lowest-visit category. This suggests that treatment protocols and follow-up schedules should be tailored by illness type rather than applied uniformly.

Insight 6: Demographic Factors Play a Secondary Role

Compared to health-related variables, demographic factors such as age, gender, and income show comparatively weaker correlations with visit frequency. This indicates that in this dataset, actual health status and condition severity matter more than demographic background when predicting healthcare utilization.

8. STATISTICAL CORRELATION SUMMARY

The table below summarizes the correlation coefficient of each major variable with the number of doctor visits. A positive value indicates that the variable increases along with visits; a negative value indicates an inverse relationship.

Variable	Correlation with Visits	Relationship Strength
Reduced Activity Days	Strong Positive	Strong
Health Score	Strong Negative	Strong
Long-term Chronic Condition	Moderate Positive	Moderate
Non-chronic Condition	Moderate Positive	Moderate
Illness Type	Moderate Positive	Moderate
Age	Weak Positive	Weak
Income	Weak Negative	Weak
Gender	Weak	Weak

9. CONCLUSION

This analysis of 5,191 patient records reveals that health-related factors — particularly chronic conditions, self-reported health score, and reduced activity days — are far more influential in determining doctor visit frequency than demographic or socio-economic factors such as age, gender, or income.

Patients suffering from chronic illnesses, reporting poor health, or experiencing extended periods of reduced activity form a clearly identifiable high-need group that accounts for a disproportionate share of total doctor visits. Recognizing and proactively managing this group could significantly improve healthcare efficiency and patient outcomes.

Overall, the findings of this project demonstrate the value of data analytics in healthcare: by systematically examining patient data, it becomes possible to identify at-risk populations, anticipate healthcare demand, and design more targeted, efficient care strategies.

10. RECOMMENDATIONS

- Implement targeted health monitoring and follow-up programs specifically for patients with long-term chronic conditions.
- Use self-reported health scores as an early warning system to flag patients likely to require more frequent care.
- Design illness-specific treatment and visit-scheduling protocols instead of a one-size-fits-all approach.
- Track reduced-activity days as a practical, easy-to-collect indicator for identifying patients who may need additional support.

- Improve healthcare access and outreach for patients on free/subsidized insurance schemes, who may have unmet health needs.
- Consider building a simple predictive model in future work to flag high-risk patients automatically based on the variables identified in this study.

— *End of Report* —